CENTRAL ENGINEERING CONSULTANCY BUREAU

Attendance System
සම්පූර්ණ සිංහල Guide

Dashboard, Attendance, OT, Reports, Voucher, Login, Database සහ Coding Structure

Project Path: `C:\github\attendance-system`
Prepared for: System understanding, handover, learning and future development
Date: 2026-07-01

1. හැඳින්වීම

මෙම guide එක CECB Attendance System එක සම්පූර්ණයෙන් තේරුම් ගන්න ලියපු ලේඛනයකි. මෙහිදී system එකේ business purpose, database connection, backend API, frontend React UI, attendance logic, OT calculation, reports, voucher generation, role-based login සහ future maintenance ගැන සරල සිංහලෙන් පැහැදිලි කරයි.

මෙය IT basic දැනුමක් තියෙන කෙනෙකුටත් system එකේ coding structure සහ data flow එක step-by-step ඉගෙන ගන්න පුළුවන් විදිහට සකස් කර ඇත.

1.1 මේ system එකෙන් කරන ප්‍රධාන වැඩ

- Employee attendance punch data බලනවා.
- CECB ERP employee details සහ workspace details connect කරනවා.
- Leave/Schedule DB එකෙන් in/out schedule ගන්නවා.
- Daily dashboard, all attendance view, analytics, AGM heatmap වගේ UI screens දෙනවා.
- Late, absent, present, short leave, half day status calculate කරනවා.
- OT summary calculate කරලා worked OT සහ payable OT පෙන්වනවා.
- OT voucher PDF/print layout generate කරනවා.
- Admin සහ Employee role-based login support කරනවා.
- Long-term data correctness සඳහා snapshots/history tables add කරලා තියෙනවා.

1.2 වැඩ ආරම්භය සිට මේ දක්වා සාරාංශය

පළමුව CECB dashboard එක check කරලා late-covered status issue එක, විශේෂයෙන් පළමු දින 2 සඳහා status correction, dashboard data display සහ attendance status logic බලලා correct කරන ලදි. ඒකෙන් පස්සේ full Attendance System එක build/extend කරන්න පටන් ගත්තා.

පසුව fingerprint attendance data, ERP employee data, workspace data, leave/schedule data එක system එකට connect කරලා reports සහ analytics develop කළා. OT side එකේ 22-16 range/monthly OT, late OT/early OT, worked OT, payable OT වගේ calculations check කරලා report and voucher side එකත් align කළා.

ඊට පස්සේ OT voucher PDF layout එක manual voucher/reference PDF වලට compare කරලා header spacing, footer spacing, table gap, signature panel වගේ layout fixes දාලා තියෙනවා. Reports loading smooth කිරීම, future month message handling, Pay Rule column removal, month/year selection වගේ UI improvements ද කළා.

අවසානයේ employee login, role-based UI, user management, database foundation, employee-user mapping, workspace history, monthly attendance snapshots සහ monthly OT snapshots add කරලා system එක long-term maintainable direction එකට ගෙන ගියා.

2. Main Technologies

Technology	මෙම system එකේ භාවිතය
React.js	Frontend UI: Dashboard, Reports, Login, Analytics, Employee screens.
ASP.NET Core Web API / .NET 8	Backend API: data read, attendance logic, report logic, authentication.
Microsoft SQL Server	ERP DB, fingerprint DB, leave/schedule DB, AttendanceSystemDB.
Entity Framework Core	C# code එකෙන් SQL Server tables query කරන ORM layer.
JWT Bearer Authentication	Login token based security. Admin/Employee role identify කරනවා.
BCrypt	User password secure hash ලෙස save කරනවා.
Recharts	React charts/graphs draw කරන library.
CSS / Utility Classes	UI styling, report layout, printable PDF-like pages.
PowerShell, dotnet CLI, npm	Development commands run කරන්න.

2.1 Recharts කියන්නේ මොකක්ද?

Recharts කියන්නේ React app එකක chart/graph draw කරන්න use කරන library එකකි. මෙම system එකේ work hours bar chart, attendance trend, present/late/absent breakdown, analytics charts වගේ UI parts වල Recharts use කරයි.

```
Data array -> Recharts component -> SVG chart
```

3. Project Folder Structure

Folder/File	Purpose
`AttendanceSystem.API`	Backend .NET API project.
`AttendanceSystem.API\Controllers`	API endpoints. Frontend එක call කරන routes.
`AttendanceSystem.API\Repository`	Main attendance/report calculation logic.
`AttendanceSystem.API\DB`	Database contexts සහ models.
`AttendanceSystem.API\BL`	Business logic layer. AuthBL login/user creation logic.
`AttendanceSystem_UI`	Frontend React app.
`AttendanceSystem_UI\src\components`	Dashboard, Reports, Analytics, Login, Employee UI components.
`AttendanceSystem_UI\src\config`	API client, auth token service, utility functions.

4. System Architecture

මෙම system එක layers කිහිපයකින් තියෙනවා. User browser එකෙන් React frontend එක open කරනවා. React frontend එක backend API endpoints call කරනවා. Backend API එක SQL Server databases වලට EF Core හරහා connect වෙලා data read/write කරනවා.

```
User Browser
   |
   v
React Frontend
   |
   v
ASP.NET Core API
   |
   +--> AttendanceERP DB        (Fingerprint punch records)
   +--> CECB_ERP / ServerERP    (Employee, designation, workspace)
   +--> LeaveDb                 (In/Out schedules)
   +--> AttendanceSystemDB      (Users, rules, snapshots, history)
```

4.1 Data Flow Example

1. User dashboard එක open කරනවා.
2. React component එක `attendanceApi.today()` වගේ API function එක call කරනවා.
3. Backend `AttendanceController` endpoint එක request එක receive කරනවා.
4. `AttendanceRepository` fingerprint DB + ERP DB + Leave DB data combine කරනවා.
5. Status calculate කරලා JSON response එක frontend එකට return කරනවා.
6. React UI table/card/chart එක update කරනවා.

5. Database Sources

Database	Context/File	Purpose
AttendanceERP	`AttendanceERPDbContext`	Fingerprint punch data. `FPDataset`, `FPuserlist`.
ServerERP / CECB_ERP	`ERPDBContext`	Employee master, designation, workspace hierarchy.
LeaveDb	`LeaveDbContext`	Employee in/out schedule times.
AttendanceSystemDB	`AppDbContext`	System users, rules, health, snapshots, workspace history.

5.1 AttendanceSystemDB Tables

Table	Purpose
`Users`	Login users. Username, password hash, role, EPF, active status.
`AttendanceRuleSettings`	Default in/out, late threshold, OT rounding/cap settings.
`DataSourceHealth`	Source DB connection health status.
`EmployeeScheduleSnapshots`	Latest employee schedule/workspace copied into app DB.
`EmployeeUserMappings`	User account and employee EPF/EmployeeId mapping.
`EmployeeWorkspaceHistory`	Workspace transfer history. Old reports can keep historical workspace.
`MonthlyAttendanceSnapshots`	Finalized monthly attendance summary copy.
`MonthlyOTSummarySnapshots`	Finalized monthly OT summary copy.
`ReportGenerationAudits`	Future report audit logging support.

6. Backend API Structure

6.1 Program.cs

`Program.cs` backend app එක start කරන main file එකයි. මෙහිදී services register කරනවා, DB contexts setup කරනවා, JWT authentication configure කරනවා, CORS setup කරනවා, Swagger enable කරනවා, startup වෙද්දී app DB operational tables create/update කරනවා.

- `AddDbContext<AppDbContext>` - AttendanceSystemDB
- `AddDbContext<AttendanceERPDbContext>` - Fingerprint DB
- `AddDbContext<ERPDBContext>` - ERP DB
- `AddDbContext<LeaveDbContext>` - Schedule DB
- `AddScoped<AttendanceRepository>` - main repository
- `AddScoped<AuthBL>` - auth business logic

6.2 Controllers

Controller	Main endpoints	Purpose
`AuthController`	`login`, `users`, `change-password`	Login, create users, list users, password change.
`AttendanceController`	`today`, `bydate`, `range`, `chart/*`	Daily/range attendance data and chart data.
`EmployeeController`	`GET /api/Employee`	Employee search/list for admin UI.
`ReportController`	`agm-wise`, `employee`, `ot-summary`, etc.	Reports and report data.

6.3 AttendanceRepository

`AttendanceRepository.cs` system එකේ brain එක වගේ. Attendance data fetch, employee join, schedule apply, status calculate, reports, OT, snapshots, workspace history වගේ core logic මෙහි තියෙනවා.

Important backend functions

Function	What it does
`GetTodayAsync()`	අද attendance records ගන්නවා.
`GetByDateAsync(date)`	එක් දිනක attendance records ගන්නවා.
`GetByRangeAsync(from, to, epfNo)`	Date range එකකට punch records ගන්නවා. EPF filter optional.
`GetStatusByEpfAndDateAsync()`	Employee එකක දිනක present/absent/late status calculate කරනවා.
`GetEmployeeReportAsync()`	Individual employee daily report summary generate කරනවා.
`GetAllEmployeeSummaryAsync()`	All employees attendance summary generate කරනවා.
`GetAgmWiseReportAsync()`	AGM -> DGM -> Service Unit hierarchy attendance summary.
`GetLateArrivalReportAsync()`	Late arrivals list.
`GetDailySummaryReportAsync()`	Daily present/absent/late/attendance rate summary.
`GetAttendanceRegisterAsync()`	Monthly attendance register data.
`GetOTSummaryAsync()`	OT worked hours, payable OT, daily OT records calculate කරනවා.
`RefreshScheduleCacheAsync()`	ERP/LeaveDB data refresh කරලා snapshots update කරනවා.

7. Authentication and Role-Based Access

Login process එක JWT token based. User login කළාම backend එක username/password verify කරලා JWT token එක return කරනවා. Frontend එක token එක localStorage එකේ save කරනවා. ඉන්පසු API requests වල Authorization header එකට token එක යවයි.

7.1 User Roles

Role	Access
Admin	Dashboard, all attendance, employee list, analytics, all reports, user management.
Employee	Own attendance dashboard, own reports, own OT/analytics. Admin reports blocked.

7.2 AuthBL important logic

- Password BCrypt hash එකෙන් verify වෙනවා.
- Failed login attempts 5ක් වුණොත් account temporary lock වෙනවා.
- Role allowlist: `Admin` හෝ `Employee` පමණයි.
- Employee user එකකට EPF අනිවාර්යයි.
- Same EPF එකට duplicate active user create වීම block වෙනවා.
- Token claim එකේ role, epfNo, fullName save වෙනවා.

> EPF numbers normalize කරලා use කරනවා. උදාහරණය: `123` -> `000123`. මේක fingerprint DB සහ ERP DB EPF formats compare කරන්න වැදගත්.

8. Frontend React Structure

Frontend app එක React components වලින් build කරලා තියෙනවා. User screen එකක action එකක් කරනකොට component state update වෙනවා, API call වෙනවා, response එකට අනුව table/chart/cards update වෙනවා.

8.1 Important UI components

Component	Purpose
`App.js`	Routing and login state. Admin/employee route guard.
`Sidebar.jsx`	Navigation menu. Admin-only menu items hide/show.
`Login.jsx`	Username/password login form.
`Dashboard.jsx`	Today attendance dashboard, KPIs, status table.
`AllAttendance.jsx`	All attendance range/table view with filters/sorting.
`Analytics.jsx`	Trend charts, AGM heatmap, period comparison.
`Reports.jsx`	AGM report, employee report, late arrivals, daily summary, all employees, OT summary, voucher.
`AttendanceRegister.jsx`	Monthly attendance register print/export.
`MyAttendance.jsx`	Employee personal dashboard and analytics.
`UserManagement.jsx`	Admin user create/list/toggle active UI.

8.2 API client

`src/config/apiClient.js` file එක frontend API calls centralize කරනවා. මෙහිදී token header attach කරනවා, auth expired නම් logout event trigger කරනවා, query string helper use කරනවා.

API object	Purpose
`attendanceApi`	Today/bydate/range/source-status/chart data.
`hrApi`	Employee list/search.
`reportApi`	Reports: AGM, employee, late, daily summary, OT, etc.
`authApi`	User management: get users, create user, toggle user.

9. Attendance Status Logic

Attendance status calculate කිරීමේදී punch data, scheduled start/end time, holiday/weekend status, source sync status යන දේවල් consider කරනවා.

9.1 Common statuses

Status	Meaning
OnTime	Schedule start time අනුව late threshold එකට කලින් check-in.
Late	Check-in scheduled start එකට පසු late threshold තුළ/ඉක්මවා.
HalfShortLeave	Late minutes configured threshold එක අනුව intermediate category.
ShortLeave	Late minutes වැඩි category.
HalfDay	Very late / half-day condition.
Absent	Working day එකක punch record නොමැතිනම්.
NotSynced	Fingerprint/source data day එකට properly sync නැති නම්.
Weekend/Holiday	Non-working day. Attendance/OT rules වෙනස් විය හැක.

9.2 Schedule logic

Employee schedule එක LeaveDb `InOutTimes` table එකෙන් ගන්නවා. Schedule නොමැති නම් default rule settings use වෙනවා. Default values current initializer එකෙන්:

- Default In: 08:30
- Default Out: 16:15
- Late threshold: 30 minutes
- Half Short Leave threshold: 45 minutes
- Short Leave threshold: 90 minutes

9.3 Dashboard correction history

Dashboard side එකේ initial correction වලදී late-covered status issue එක සහ first 2 days status problem එක check කරලා accurate status display කරගන්න improvements කළා. Dashboard එකේ data source, late status calculation, table row status display වගේ parts align කිරීම මෙම stage එකේ සිදු වුණා.

10. OT Logic

OT summary report එක worked OT සහ payable OT දෙක පෙන්වයි. Worked OT කියන්නේ actual worked overtime duration. Payable OT කියන්නේ rules අනුව ගෙවිය හැකි OT amount/hours.

10.1 Worked OT vs Payable OT

Column	Meaning
Worked OT	Employee actual වැඩ කරපු overtime hours. Morning OT + Evening OT.
Payable OT	Payment rules අනුව ගෙවිය හැකි OT hours. Engineer pay category වගේ rules බලයි.

10.2 OT calculation steps

1. Employee punch record එකේ check-in/check-out ගන්නවා.
2. Employee schedule start/end time resolve කරනවා.
3. Morning OT: scheduled start ට කලින් valid early work.
4. Evening OT: scheduled end ට පස්සේ valid extra work.
5. Weekend/Holiday නම් non-working day OT rules apply කරනවා.
6. Rounding minutes and OT cap rules apply කරනවා.
7. Total OT hours calculate කරනවා.
8. Engineer category නම් payable OT special rule apply කරනවා.

10.3 OT voucher

OT voucher PDF/print layout එක Reports UI තුළ generate කරයි. Voucher layout එක manual/reference voucher PDF වලට compare කරලා header alignment, footer bottom spacing, table-to-signature spacing, line 7 to Additional Hours gap වගේ details fix කර ඇත.

Voucher generate කරනකොට selected month range එකට load වූ report data use කරනවා. Data mismatch avoid කරන්න selected month වෙනස් කළ පසු report generate නොකර export/voucher button use කිරීම guard කරලා තියෙනවා.

11. Reports Module

Reports module එක admin සහ employee දෙපාර්ශවයට වෙනස් permission සමඟ available වේ. Admin all reports බලනවා. Employee තමන්ගේ EPF එකට අදාල reports පමණක් බලනවා.

11.1 Report types

Report	Purpose
AGM-wise Attendance Report	AGM -> DGM -> Service Unit hierarchy attendance statistics.
Employee Report	One employee daily attendance details for a date range.
Monthly Sheet	Monthly per-day status sheet.
EPF Status Lookup	Single employee/day status lookup.
Late Arrivals Report	Late employees list for date range.
Daily Summary	Daily total registered, present, absent, late, rate.
All Employee Summary	All employees attendance summary by date range.
OT Summary	Month/year based OT summary and voucher data.
Attendance Register	Physical attendance register print/export.

11.2 Month/year OT report change

OT Summary එක මුලින් date range + voucher month එකක් තිබුණු නිසා confusion එකක් තිබුණා. ඒක month/year selection flow එකකට improve කළා. Date range option වෙනුවට user තෝරන්නේ year සහ month. System එක ඒ month එකේ first day සිට last day දක්වා range build කරනවා.

> Future month එකක් select කළොත් error-like red message නොව, friendly info message එකක් පෙන්වයි. Generate button disable වෙනවා.

11.3 Exports

- CSV export: table rows CSV file එකක් ලෙස download.
- PDF/Print export: browser print window එකෙන් PDF save කළ හැක.
- Voucher PDF: OT voucher layout එක print/PDF ලෙස generate.

12. Analytics and Heatmap

Analytics page එක attendance trends, AGM heatmap, period comparison වගේ decision support UI එකක්. Heatmap color එක attendance percentage අනුව වෙනස් වෙනවා. Legend boxes date range data loaded වූ පසු පමණක් පෙන්වයි.

12.1 Heatmap meaning

Color/Range	Meaning
Excellent	High attendance percentage.
Good	Acceptable attendance level.
Watch	Attention needed soon.
Low / Needs attention	Management attention required.

13. Employee Login Experience

Employee user login වූ විට admin dashboard වෙනුවට personal attendance view එකක් පෙන්වයි. Employeeට තමන්ගේ attendance, status breakdown, work hours chart, OT/report data වගේ own data පමණක් access කළ හැක.

13.1 Employee dashboard features

- Own EPF based date range attendance load.
- Present, late, absent, attendance percentage KPIs.
- Work hours bar chart.
- Status breakdown chart/cards.
- Own report/OT access.

13.2 Admin user management

Admin user can open Users page and create Employee/Admin users. Employee user create කරනකොට EPF select කළ හැක. EPF selected වූ විට employee full name and username auto-fill support ඇත.

14. Snapshot and History Concept

Snapshot කියන්නේ ඒ වෙලාවේ තිබුණු data copy එකක්. Live ERP data වෙනස් වුණත් old report එක correct historical data පෙන්වන්න මේක වැදගත්.

14.1 Example

May 2026 report generate කළ වෙලාවේ employee එක Colombo service unit එකේ සිටිය හැක. July වල Kandy service unit එකට transfer වුණොත්, May report නැවත බලද්දී May data Colombo ලෙසම තිබිය යුතුයි. ඒ නිසා workspace history/snapshot data save කිරීම අවශ්‍යයි.

14.2 Snapshot tables

- `EmployeeScheduleSnapshots` - latest schedule/workspace copy.
- `EmployeeWorkspaceHistory` - transfer history with effective dates.
- `MonthlyAttendanceSnapshots` - finalized attendance summary.
- `MonthlyOTSummarySnapshots` - finalized OT summary.

15. Coding Concepts for Basic IT Learner

15.1 React component කියන්නේ මොකක්ද?

React component කියන්නේ screen එකේ එක් කොටසක් draw කරන JavaScript function එකක්. උදාහරණයට `Dashboard()` component එක dashboard screen එක draw කරනවා.

```
export default function Dashboard() {
  // state, API calls, render UI
  return (...);
}
```

15.2 State කියන්නේ මොකක්ද?

State කියන්නේ component එකේ current data. උදාහරණයට loading true/false, selected date, report data, table filter වගේ දේවල් state වේ.

15.3 API call කියන්නේ මොකක්ද?

Frontend එක backend එකෙන් data ඉල්ලන request එක. උදාහරණයට dashboard එක today attendance data ඉල්ලයි.

```
const rows = await attendanceApi.today();
```

15.4 Controller කියන්නේ මොකක්ද?

Backend controller කියන්නේ API route එක receive කරන C# class එකක්. Frontend request එක route එකට එනවා, controller එක repository/business logic call කරලා result return කරනවා.

15.5 DbContext කියන්නේ මොකක්ද?

DbContext කියන්නේ EF Core හරහා database tables code objects ලෙස handle කරන class එකක්. උදාහරණයට `AppDbContext.Users` කියන්නේ Users table එක query කරන්න use කරන object එකයි.

15.6 DTO කියන්නේ මොකක්ද?

DTO (Data Transfer Object) කියන්නේ API response/request shape එක define කරන class එකක්. Backend එක frontend එකට data යවන format එක DTO වලින් පැහැදිලි කරයි.

16. Important Commands

16.1 Backend run

```
cd C:\github\attendance-system\AttendanceSystem.API
dotnet restore
dotnet build
dotnet run
```

> Command එක `dotnet run`. `dotnetrun`, `dot net run`, `do net run` වැරදියි.

16.2 Frontend run

```
cd C:\github\attendance-system\AttendanceSystem_UI
npm install
npm start
```

16.3 Build verify

```
cd C:\github\attendance-system\AttendanceSystem.API
dotnet build --no-restore -o build-verify

cd C:\github\attendance-system\AttendanceSystem_UI
npm run build
```

17. Common Errors and Fixes

Error	Reason	Fix
`dotnetrun not recognized`	Command spacing wrong.	Use `dotnet run`.
`Failed to fetch`	API not running, wrong port, CORS/API URL issue.	Start backend, check frontend API base URL.
Slow report loading	ERP/LeaveDB joins, large date range, cache cold.	Use month filters, cache warmup, snapshot support.
Debug exe locked	Existing API process running.	Stop old process or build to `build-verify`.
EF IntelliSense red errors	VS Code language server/cache issue.	Run restore/build, reload VS Code, restart C# language server.

18. Data Accuracy Rules

- Reports should use correct loaded date/month range.
- OT voucher should use the same loaded month data, not stale selected month.
- Employee self-access should compare normalized EPF.

- Future month reports should show friendly info and block generation.
- Historical workspace should be stored in workspace history/snapshot tables.
- Snapshot writes should never break live report generation, but should log warnings.

19. Future Improvements

- `ReportGenerationAudits` table එකට actual report generation audit rows save කිරීම.
- Monthly finalized reports snapshot-first mode එකට gradually move කිරීම.
- Trainee Allowance Voucher separate module එකක් add කිරීම.
- Admin UI එකෙන් attendance rules edit කිරීම.
- Data source health dashboard එක improve කිරීම.
- Automated tests add කිරීම: OT calculation, late status, auth role access.

19.1 Trainee Allowance Voucher note

User provided image එකෙන් පේන්නේ Trainee Allowance Voucher format එකක්. එය OT voucher එකෙන් වෙනස් report type එකක්. Future module එකක් ලෙස attendance days, time-in/time-out, total days, allowance amount, signatures සමඟ separate voucher generate කළ හැක.

20. Final Summary

මෙම Attendance System එක React frontend + ASP.NET Core backend + SQL Server databases මත build කර ඇත. System එක fingerprint data, ERP employee data, LeaveDB schedule data combine කරලා attendance, late/absent, reports, OT and voucher logic handle කරයි. Role-based login, employee personal dashboard, admin management, report exports, snapshots/history foundation ඇතුළත් නිසා system එක long-term maintainable සහ expandable direction එකකට ගෙන ගොස් ඇත.

> Basic learner සඳහා මතක තියාගන්න: Frontend UI එක React වලින් data ඉල්ලනවා. Backend API එක C# වලින් logic process කරනවා. SQL Server databases data store/read කරනවා. Reports/Vouchers මේ processed data වලින් generate වෙනවා.